

Data Science & ML Intern Assignment (GoalFi, IIT Kanpur)

Project Title:

Predicting Stock Price Movements Using Machine Learning Techniques

Overview:

This project is designed to assess how you handle real-world data, design end-to-end ML pipelines, and communicate insights through data-driven analysis. The goal is to predict next-day stock price movements (Up/Down) for a selected Indian stock using historical market data.

We're not expecting a production-grade trading model — rather, we want to see how you think, structure your work, and explain your findings clearly.

Assignment Details

1. Data Collection & Preparation

Use any freely available Indian equity dataset, such as NIFTY 50 constituent stock data (OHLCV — Open, High, Low, Close, Volume) covering the last 2 years.

Data can be fetched using sources like Yahoo Finance, Alpha Vantage, or NSE APIs.

Perform data cleaning: handle missing values, outliers, and ensure data is in chronological order.

2. Feature Engineering

Generate at least three technical indicators (examples: RSI, Moving Averages, MACD, Bollinger Bands, etc.) as features.

Feel free to include additional features like lagged returns, volatility, or volume ratios if relevant.

Clearly explain why you chose each feature and what it represents in market behavior.

3. Model Development

Define your target variable as the next-day price direction (1 = Up, 0 = Down).

Split the data into training and testing sets using a time-based split.

Train at least two ML models, such as:

Logistic Regression

Random Forest

XGBoost

(You may explore others if you wish — SVM, LightGBM, etc.)

Compare the performance of your models.

4. Model Evaluation

Evaluate your models using appropriate classification metrics:

Accuracy

Precision, Recall, F1-Score

Confusion Matrix

Include visualizations (e.g., confusion matrix heatmap, feature importance chart, ROC curve) to make your results interpretable.

5. Insights & Visualization

Highlight the most predictive features from your analysis.

Use visualizations (Matplotlib, Seaborn, or Plotly) to:

Show feature importance

Compare actual vs. predicted signals

Illustrate model performance

6. Reporting

Document your approach and findings briefly:

Key steps and methodology

Model performance summary

Observations and insights

Limitations and potential improvements

Bonus (Optional)

If you'd like to go a step further, you can:

Deploy your ML pipeline as a simple Flask web app or an interactive notebook dashboard (e.g., Streamlit, Dash).

This will demonstrate your ability to translate ML models into usable applications.

Advanced Extension (Optional for Enthusiasts)

For candidates interested in research and advanced ML applications:

Outline how your system could evolve into a GenAI / Agentic AI or Reinforcement Learning (RL)-driven architecture for automated trading or portfolio management.

You may describe or prototype:

State: Features or signals the model observes (e.g., recent returns, volatility, volume indicators)

Actions: Possible trading or allocation decisions

Reward: Objective function (e.g., Sharpe ratio, profit, or risk-adjusted return)

Training: How the agent learns from market feedback over time

A conceptual framework or pseudocode is perfectly fine — full implementation is not required.

What We're Looking For

Good understanding of data preprocessing and feature engineering

Sound application of ML algorithms and model evaluation

Clean, readable, and well-commented code (Python preferred)

Effective data visualization and presentation of results

Clear communication of your reasoning, assumptions, and findings

(Optional) Curiosity and innovation in extending the model

Submission Format

Please submit:

A Jupyter Notebook (.ipynb) containing your code, analysis, and visualizations

A short report (PDF or Markdown) summarizing your workflow, results, and conclusions

File naming convention:

YourName_DataScienceAssignment.ipynb

YourName_Report.pdf